Week 12 — Non-Linear Time Series: ARCH & GARCH

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The big picture. For the changing volatility of financial returns (volatility clustering) we model the conditional variance: ARCH/GARCH are uncorrelated yet dependent.

Linear models (AR, MA, ARMA) describe the conditional *mean* and assume a constant innovation variance. Financial log-returns break both habits: they are nearly *uncorrelated* (a conditional-mean model is almost useless), yet their *squares* are strongly dependent — big moves cluster with big moves, calm with calm. This **volatility clustering** plus heavy (**fat**) tails calls for a model of the *conditional variance*. ARCH/GARCH keep the series mean-zero white noise while letting σ_t^2 evolve with past squared returns (and, for GARCH, past variances). Throughout, $\mathcal{F}_t$ is the information up to time t, and $\{\varepsilon_t\}$ is white noise with $\mathbb{E}[\varepsilon_t] = 0$, $\mathrm{Var}(\varepsilon_t) = 1$, independent of $\mathcal{F}_{t-1}$.

Key definitions

Definition — Log-returns & volatility clustering

For prices $\{X_t\}$, the **log-returns** are $r_t = \log(X_t/X_{t-1}) = \log X_t - \log X_{t-1}$ (relative price changes). Empirically: (i) the ACF/PACF of r_t looks like *white noise*; (ii) the ACF/PACF of r_t^2 *decays slowly* $\Rightarrow$ the conditional variance changes over time; (iii) **volatility clustering** (large changes follow large, small follow small); (iv) **fat tails**, kurtosis > 3 (a QQ-plot against $\mathcal{N}(0,1)$ bends at the ends).

Definition — ARCH(p) model

$\{r_t\}$ is **AutoRegressive Conditionally Heteroscedastic** of order p if

$$r_t = \sigma_t \varepsilon_t, \qquad \sigma_t^2 = \alpha_0 + \sum_{j=1}^{p} \alpha_j\, r_{t-j}^2,$$

with $\alpha_0 > 0$, $\alpha_p > 0$, $\alpha_j \geq 0$. The volatility σ_t is $\mathcal{F}_{t-1}$-measurable (known one step ahead), so $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0$ and $\mathrm{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$. The case $p = 1$ is $\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2$.

Definition — GARCH(p,q) model

$\{r_t\}$ is **Generalised ARCH** if $r_t = \sigma_t \varepsilon_t$ with

$$\sigma_t^2 = \alpha_0 + \sum_{j=1}^{p} \alpha_j\, r_{t-j}^2 + \sum_{j=1}^{q} \beta_j\, \sigma_{t-j}^2,$$

$\alpha_0 > 0$, $\alpha_p, \beta_q > 0$, other $\alpha_j, \beta_j \geq 0$. Feeding back past *variances* σ_{t-j}^2 makes volatility persistent far more cheaply than a large-p ARCH. A GARCH($p, 0$) is exactly an ARCH(p); again $\mathrm{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$.

Definition — Standardised residuals (diagnostics)

After fitting, $\tilde{e}_t = r_t/\hat{\sigma}_t$. If the model is right these look i.i.d. $(0,1)$: the ACF/PACF of $\tilde{e}_t$ *and* of the squares $\tilde{e}_t^2$ should resemble white noise, and a QQ-plot against the assumed innovation law should be roughly straight. One also overlays the in-sample bands $\pm 1.96\,\hat{\sigma}_t$ on r_t (Gaussian innovations).

Definition — ARMA–GARCH model

Since a GARCH process is itself white noise, it can be the innovation of an ARMA: with r_t a GARCH process,

$$X_t = \sum_{j=1}^{p} \phi_j X_{t-j} - \sum_{k=0}^{q} \theta_k\, r_{t-k}, \qquad \theta_0 = -1,$$

the ϕ_j, θ_k satisfying the usual ARMA conditions. This models a conditional mean (ARMA) and a time-varying conditional variance (GARCH) at once — just be careful with the likelihood, as errors are no longer i.i.d. Gaussian.

Why feed back σ^2_{t-1}?

In ARCH(1) the variance depends only on the *single* most recent r^2_{t-1}: one quiet day collapses σ_t^2 back toward α_0, so the model cannot *sustain* a high-volatility regime. GARCH adds the smooth term $\beta_1\sigma^2_{t-1}$: yesterday's variance estimate is mostly carried forward, so volatility is **sticky**. Fitted GARCH$(1,1)$ typically has $\alpha_1 + \beta_1$ close to 1 — long, persistent memory (still geometric decay, but very slow).

Key results & formulas

Proposition — moments of ARCH(1)

For ARCH(1) with $\alpha_1 < 1$:

$$\begin{aligned} \mathbb{E}[r_t \mid \mathcal{F}_{t-1}] &= 0, \quad \operatorname{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2 = \alpha_0 + \alpha_1 r^2_{t-1}, \\ \mathbb{E}[r_t] &= 0, \qquad\quad \operatorname{Var}(r_t) = \frac{\alpha_0}{1-\alpha_1}. \end{aligned}$$

Idea: σ_t is known given $\mathcal{F}_{t-1}$ and ε_t is independent of it, so $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = \sigma_t \mathbb{E}[\varepsilon_t] = 0$ and $\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$; take unconditional expectations and use stationarity $\operatorname{Var}(r_t) = \operatorname{Var}(r_{t-1})$. Here $\alpha_1 < 1$ is *necessary and (for ARCH(1)) sufficient* for a finite, positive variance.

Proposition — (G)ARCH is white noise (uncorrelated but dependent)

Any ARCH(p) or GARCH(p,q) is mean-zero and serially uncorrelated: for $\tau \neq 0$,

$$\mathbb{E}[r_t] = 0, \qquad \operatorname{Cov}(r_{t+\tau}, r_t) = 0, \qquad \operatorname{Corr}(r_{t+\tau}, r_t) = 0.$$

Idea: for $\tau > 0$, both r_t and $\sigma_{t+\tau}$ are $\mathcal{F}_{t+\tau-1}$-measurable, so $\mathbb{E}[r_{t+\tau} r_t] = \mathbb{E}\big[r_t \sigma_{t+\tau}\, \mathbb{E}[\varepsilon_{t+\tau} \mid \mathcal{F}_{t+\tau-1}]\big] = 0$. The qualitative punchline: (G)ARCH leaves the *linear* correlation flat (white noise) yet creates strong *nonlinear* dependence in the squares. Mean and variance modelling decouple.

Theorem — Isserlis (Wick formula) & ARCH(1) kurtosis

For a zero-mean Gaussian vector, $\mathbb{E}[X_1X_2X_3X_4] = \mathbb{E}[X_1X_2]\mathbb{E}[X_3X_4] + \mathbb{E}[X_1X_3]\mathbb{E}[X_2X_4] + \mathbb{E}[X_1X_4]\mathbb{E}[X_2X_3]$; in particular $\mathbb{E}[\varepsilon_t^4] = 3$. Hence, for ARCH(1) with Gaussian noise, the fourth moment is finite **iff** $\alpha_1^2 < \frac{1}{3}$, and

$$\mathbb{E}[r_t^4] = \frac{3\alpha_0^2(1+\alpha_1)}{(1-\alpha_1)(1-3\alpha_1^2)}, \qquad \kappa = \frac{\mathbb{E}[r_t^4]}{\operatorname{Var}(r_t)^2} = \frac{3(1-\alpha_1^2)}{1-3\alpha_1^2} > 3 \;\; (\alpha_1 > 0).$$

So ARCH *generates* excess kurtosis (fat tails) on its own, even with Gaussian ε_t.

Proposition — ACF of the squares of ARCH(1)

For ARCH(1) with $0 < \alpha_1 < 1/\sqrt{3}$,

$$\mathrm{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}, \qquad \tau \in \mathbb{Z}:$$

the autocorrelation of the *squares* decays **geometrically**. *Idea:* condition on $\mathcal{F}_{t+\tau-1}$ to peel one lag, giving the recursion $\mathrm{Cov}(r_{t+\tau}^2, r_t^2) = \alpha_1 \mathrm{Cov}(r_{t+\tau-1}^2, r_t^2)$ (constant terms cancel via $\mathbb{E}[r_t^2] = \alpha_0/(1-\alpha_1)$). This is often *too fast* for real data — motivating GARCH.

Proposition — GARCH(p,q) stationarity, variance & 4th moment

A GARCH(p,q) has a weakly stationary solution **iff** $\sum_j \alpha_j + \sum_j \beta_j < 1$, and then

$$\mathrm{Var}(r_t) = \frac{\alpha_0}{1 - \sum_{j=1}^{p} \alpha_j - \sum_{j=1}^{q} \beta_j}.$$

For GARCH$(1,1)$ with Gaussian noise the fourth moment exists **iff** $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1$ (the analogue of $\alpha_1^2 < 1/3$). *Note:* $\sum \alpha + \sum \beta < 1$ is also sufficient for strict stationarity; the boundary $\sum \alpha + \sum \beta = 1$ (Gaussian noise) is strictly — not weakly — stationary: the IGARCH case.

Proposition — conditional-Gaussian MLE

Conditioning on r_1 (so no marginal law for the initial value), the likelihood factorises and each $r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, \sigma_t^2)$. Up to constants,

$$\log L_c = -\tfrac{1}{2} \sum_t \left[\log \sigma_t^2 + \frac{r_t^2}{\sigma_t^2} \right],$$

maximised numerically (with σ_t^2 from the model recursion) subject to positivity and $\sum \alpha + \sum \beta < 1$. The per-time variance σ_t^2 is what is fitted, not a constant.

Intuition

White noise $\neq$ no structure

The headline paradox: $\mathrm{Corr}(r_{t+\tau}, r_t) = 0$ (uncorrelated, so linear forecasting of r_t is hopeless) *but* $\mathrm{Corr}(r_{t+\tau}^2, r_t^2) \neq 0$ (the magnitudes are predictable). ARCH/GARCH live exactly here — flat ACF, dependent squares. That is also why they slot in cleanly as ARMA innovations: they do not disturb the correlation structure.

Two nested thresholds — don't confuse them

$\alpha_1 < 1$ buys a finite *variance*; the strictly tighter $\alpha_1^2 < \frac{1}{3}$ (i.e. $\alpha_1 < 1/\sqrt{3} \approx 0.577$) buys a finite *fourth* moment, needed before any kurtosis or ACF-of-squares formula is valid. As $\alpha_1 \uparrow 1/\sqrt{3}$, $\kappa \to \infty$: more ARCH effect $\Rightarrow$ heavier tails. (GARCH$(1,1)$ analogue: $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1$.)

GARCH(1,1) $\approx$ ARCH(∞), but parsimonious

Iterating $\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \beta_1 \sigma_{t-1}^2$ gives $\sigma_t^2 = \frac{\alpha_0}{1-\beta_1} + \alpha_1 \sum_{k\geq 0} \beta_1^k r_{t-1-k}^2$ — an ARCH(∞) with geometrically decaying weights $\alpha_1 \beta_1^k$. Three numbers $(\alpha_0, \alpha_1, \beta_1)$ encode the whole slow tail: α_1 = immediate shock impact, β_1 = persistence/decay rate. An ARCH(p) would need many α_j to mimic this.

Reading the (G)ARCH likelihood

Each factor is a normal with its *own* variance σ_t^2. The log-likelihood trades off "don't make σ_t^2 needlessly large" (the $\log \sigma_t^2$ penalty) against "explain a big r_t^2 with a big σ_t^2" (the r_t^2/σ_t^2 term). That balancing act *is* volatility tracking.

Pros & cons

Pros: capture volatility clusters and time-varying variance; reproduce fat tails (large kurtosis); GARCH gives persistent volatility cheaply; they don't disturb the white-noise correlation. **Cons:** positive and negative shocks act *symmetrically* (no leverage effect); volatility dependence decays only *geometrically*, often too fast for real markets. Extensions (EGARCH, GJR-GARCH, long-memory GARCH) fix these but are out of scope.

GARCH(1,1) sanity checks (S&P 500 fit)

Fitted: $\alpha_0 = 3.686 \times 10^{-6}$, $\alpha_1 = 0.1713$, $\beta_1 = 0.7897$. *Stationary?* $\alpha_1 + \beta_1 = 0.961 < 1$ ✓, with $\mathrm{Var}(r_t) = \alpha_0/(1 - 0.961) \approx 9.45 \times 10^{-5}$, i.e. daily volatility $\sqrt{\cdot} \approx 0.97\%$. *Fourth moment?* $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 \approx 0.982 < 1$ ✓(only just — very heavy tails). The persistence $\alpha_1 + \beta_1 = 0.961$ near 1 is the hallmark of financial GARCH: long quiet/turbulent regimes, yet still stationary.

Key takeaways

Key takeaways — the week's reflexes

- **Model the variance, not the mean.** $r_t = \sigma_t \varepsilon_t$; ARCH: $\sigma_t^2 = \alpha_0 + \sum \alpha_j r_{t-j}^2$; GARCH adds $\sum \beta_j \sigma_{t-j}^2$. Always $\mathrm{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$.
- **White noise but dependent:** $\mathrm{Corr}(r_{t+\tau}, r_t) = 0$ yet $\mathrm{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}$ (ARCH(1)) $\neq 0$. Diagnose with the ACF of *squared* residuals.
- **Long-run variance / stationarity in one move:** take unconditional expectations of the σ_t^2 recursion, set $\mathbb{E}[r_{t-j}^2] = \mathbb{E}[\sigma_{t-j}^2] = \mathrm{Var}(r_t)$, solve $\mathrm{Var}(r_t) = \alpha_0/(1 - \sum \alpha - \sum \beta)$; positivity $\Leftrightarrow \sum \alpha + \sum \beta < 1$.
- **Two thresholds:** $\alpha_1 < 1$ (variance) vs. $\alpha_1^2 < 1/3$ (kurtosis); GARCH(1,1) 4th moment: $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1$.
- **Fat tails for free:** Gaussian ε_t + ARCH $\Rightarrow \kappa > 3$.
- **Fit by MLE:** maximise $-\frac{1}{2}\sum_t [\log \sigma_t^2 + r_t^2/\sigma_t^2]$ with σ_t^2 from the recursion. GARCH plugs into ARMA as the innovation (ARMA–GARCH).